

Bases and alkalis

The acidity of vinegar (ethanoic acid) can be neutralized, or cancelled out, by adding chalk (calcium carbonate). Any substance that neutralizes acidity, such as chalk, is called a base. An alkali is a base that dissolves in water. An alkali's strength is measured by the number of hydroxide ions it forms in water. Strong alkalis, such as sodium hydroxide, are just as corrosive as strong acids.

The mixture spills out of the flask.

Testing the mixture with universal indicator solution proves that it is now neutral – the acidity has been cancelled out.

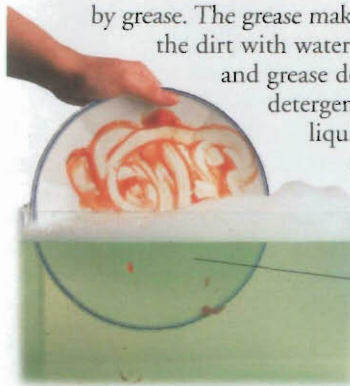
Chalk and vinegar react together and release carbon dioxide gas.

The product of the reaction is a salt called calcium ethanoate.

Soaps and detergents

Alkalis are good at dissolving oil and grease, so they are widely used in the manufacture of soaps and detergents.

Most dirt is bound to skin, clothes, or eating utensils by grease. The grease makes it difficult to remove the dirt with water alone, because water and grease do not mix. A soap or detergent, such as washing-up liquid, breaks the grease up into tiny drops and allows the water to wash away the dirt.



Once the washing-up liquid has broken down the grease, the water can wet the plate and dissolve the rest of the dirt.



Oil slicks

Accidents with oil tankers at sea can create huge oil slicks (spillages) on the water's surface. Strong detergents called dispersants may be used to break up the oil. Wildlife experts use weaker detergents, such as washing-up liquid, to clean the feathers of oil-coated seabirds. If the birds' feathers – which usually keep them warm and dry – become clogged with oil, the birds may lose their buoyancy and drown, or die of exposure to the cold.

Batteries

Acids, alkalis, and salts are electrolytes, meaning that they conduct electricity when in solution. Batteries consist of an electrolyte – usually in the form of a moist paste or liquid – between two rods or plates called electrodes. The most common battery is the dry cell, which uses the salt ammonium chloride as an electrolyte. Long-life batteries contain alkaline electrolytes, such as potassium hydroxide; car batteries have electrolytes of sulphuric acid.



Alkali industry

The main raw material in the alkali industry is brine (salt water). Sodium hydroxide, which is used to make soaps and paper, is produced from brine by electrolysis (passing electricity through it). Brine will also absorb carbon dioxide to make sodium carbonate, which is used in textile treatment, photography, and glass making.

Electrolysis of brine to make sodium hydroxide



Neutralizing acids

An alkali and an acid react together to give a neutral salt. In addition, hydroxide ions (OH^-) in the alkali combine with the acid's hydrogen ions (H^+) to produce water (H_2O). In daily life, problems of unwanted acidity are solved by adding an alkali of the appropriate strength.

Soil acidity

The pH of soil varies from area to area. Few crops grow well in highly acidic soil, because the acid dissolves vital minerals that the plants need for healthy growth and allows them to be washed away. Farmers treat acidic soil by spreading lime (calcium oxide) over their fields. This is a cheaply produced alkali made from limestone. It neutralizes the acid in the soil, making it more fertile.



Farmer liming acidic soil

Stomach powder fizzes as it reacts with lemon juice (citric acid).



Curing indigestion

The human stomach uses hydrochloric acid to break down food. Some foods cause your stomach to produce so much acid that it gives you discomfort. Stomach powders or indigestion tablets can cure this. They contain weak alkalis that neutralize the acidity, but do not harm your stomach, or react too vigorously with the acid.

Bee and wasp stings

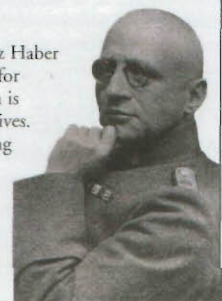
A bee sting is painful because it is acidic. Treating the sting with a weak alkali, such as soap or bicarbonate of soda, relieves the pain by neutralizing the acid. In contrast, a wasp sting is alkaline, so it can be neutralized by a weak acid, such as vinegar or lemon juice.



Bee

Fritz Haber

In 1908, the German chemist Fritz Haber (1868–1934) developed a process for making the alkali ammonia, which is used to make fertilizers and explosives. The Haber process involves reacting nitrogen from the air with hydrogen at high pressure and temperature. Haber later devised a way of making nitric acid by heating ammonia in air.



Timeline

c.600 BC The Phoenicians use alkaline wood ash to make soap.

11th century AD Arab chemists make sulphuric, nitric, and hydrochloric acids.

1780s World's first sulphuric acid factory opens in France.

1865 Ernest Solvay, a Belgian chemist, develops the first commercially successful process for making the alkali, sodium carbonate, on a large scale.



Sodium carbonate

1887 Svante Arrhenius proposes that it is hydrogen ions that give acids their special properties.

1908 Fritz Haber invents a process for making ammonia.

1909 The Danish chemist Søren Sørensen (1868–1939) devises the pH scale.

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